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Design and Implementation of a Redundant Enterprise Campus Network

Technical Documentation

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Platform	Cisco Packet Tracer
Scope	Three-floor redundant campus network
Devices	2 MLS, 6 access switches, 2 edge routers, 1 ISP router
Status	Fully configured and verified

All configurations and design decisions are original work

Abstract

This document presents the design and full configuration of a redundant enterprise campus network built in Cisco Packet Tracer. The topology models a three-floor office environment serving six organizational departments, each isolated in a dedicated VLAN with its own IP subnet.

The architecture follows the three-tier hierarchical model and integrates the following technologies: VLAN segmentation with VTP propagation, inter-VLAN routing via Switch Virtual Interfaces on dual multilayer switches, first-hop redundancy using HSRP, loop-free switching using Rapid PVST+ with optimized root bridge placement, link aggregation using a 3-link LACP EtherChannel, centralized DHCP service via relay, and northbound static routing through dual edge routers connected to a simulated ISP.

All configurations are original work and have been verified using standard Cisco `show` commands within the simulator. End-to-end DHCP operation was confirmed by capturing live host address assignments. This document serves as a technical portfolio artifact demonstrating practical skills in enterprise network design, IP addressing, redundancy planning, and Layer 2/3 configuration.

Key technologies	VLANs, 802.1Q trunking, VTP, SVI routing, HSRP, Rapid PVST+, LACP EtherChannel, DHCP relay, static routing
IP ranges used	172.16.0.0/16 (campus), 10.0.0.0/24 (core links), 198.51.100.0/24 & 203.0.113.0/24 (ISP simulation, RFC 5737)
Simulation tool	Cisco Packet Tracer

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1 Project Overview

This project presents the design and complete configuration of a redundant enterprise campus network built entirely in Cisco Packet Tracer. The topology simulates a three-floor office environment serving multiple organizational departments and integrates a structured set of enterprise networking technologies to ensure traffic segmentation, high availability, and controlled routing behavior.

The architecture follows the hierarchical campus model — access, distribution/core, and edge — covering VLAN segmentation, inter-VLAN routing via multilayer switching, gateway redundancy using HSRP, loop prevention using Rapid PVST+, link aggregation using LACP EtherChannel, centralized DHCP relay, static routing with floating backup paths, and a simulated ISP uplink via dual edge routers.

Simulation environment: Cisco Packet Tracer only. The ISP router and public server simulate external reachability inside the Packet Tracer environment and do not represent a real Internet connection. All device behaviors and protocol interactions are subject to Packet Tracer simulator constraints.

2 Network Topology

2.1 Architecture Overview

The network follows a three-tier hierarchical design composed of four functional layers:

- **Access layer:** six access switches (SW2–SW7) distributed across three floors. Each switch serves one VLAN and is dual-homed to both multilayer switches through 802.1Q trunk links, ensuring a redundant uplink at every access point.
- **Distribution/Core layer:** two multilayer switches (MLS-1 and MLS-2) providing inter-VLAN routing through SVIs, HSRP first-hop redundancy, Rapid PVST+ root bridge control, and a 3-link LACP EtherChannel between them for bandwidth aggregation and link redundancy.
- **Edge layer:** two routers (R-1 and R-2) each connected to both multilayer switches through dedicated /30 routed links, providing full-mesh redundancy at the campus boundary.
- **Public simulation layer:** one ISP router connected to a simulated public segment (203.0.113.0/24), used to verify end-to-end reachability through the routing domain.

2.2 Topology Diagram

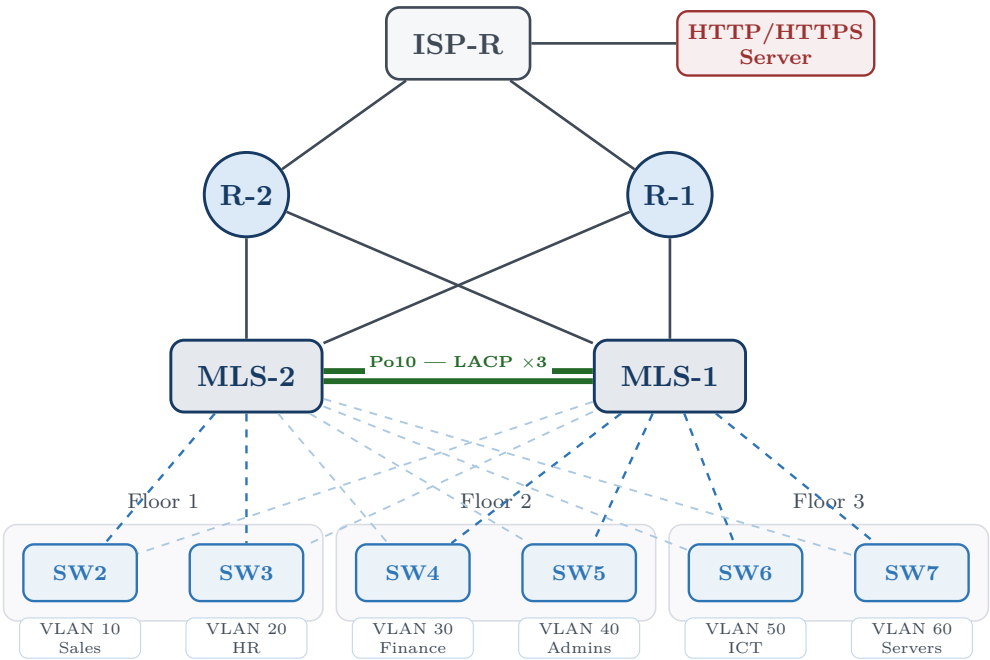


Figure 1: Logical network topology — three-floor enterprise campus

Legend: — EtherChannel (LACP ×3) - - - 802.1Q trunk (primary) - - - 802.1Q trunk (secondary/dual-home) — Routed link

2.3 Floor Distribution

Table 1: Department-to-floor mapping

Floor	Switch	VLAN	Department	Notes
1	SW2	10	Sales	User access switch
1	SW3	20	HR	User access switch
2	SW4	30	Finance	User access switch
2	SW5	40	Admins	User access switch
3	SW6	50	ICT	User access switch
3	SW7	60	Servers	Hosts the Packet Tracer DHCP server (172.16.60.4)

2.4 Traffic Flow — End-to-End Path Analysis

Understanding how traffic traverses the network validates the design intent. The following example traces a complete communication path from an internal host to the simulated public server.

Scenario: a Sales department host in VLAN 10 sends a request to the HTTP/HTTPS server at 203.0.113.10.

1. The host has obtained its IP address (e.g. 172.16.10.5/25) via DHCP and uses the HSRP virtual IP 172.16.10.3 as its default gateway.

2. The frame is sent to the MAC address of the HSRP virtual IP. MLS-1, as the active HSRP gateway for VLAN 10, receives the frame on its SVI and performs a Layer 3 routing lookup.
3. The destination (203.0.113.10) is not in any directly connected subnet. MLS-1 follows its primary default static route toward R-1 (10.0.0.2) via Gi1/0/10.
4. R-1 receives the packet on G0/2. Its routing table contains a static route toward 172.16.0.0/16 and a default route toward the ISP (198.51.100.1). R-1 forwards the packet to ISP-R.
5. ISP-R routes the packet to 203.0.113.10 on its directly connected public segment.
6. The return path follows the same chain in reverse: ISP-R forwards to R-1, which routes back toward 172.16.0.0/16 via MLS-1, which delivers the frame to the host on VLAN 10.

If MLS-1 or R-1 become unavailable, HSRP transfers the active gateway role to MLS-2 automatically, and the floating static routes direct traffic toward R-2, ensuring full continuity with no reconfiguration required on end hosts.

3 IP Addressing Plan

3.1 Campus VLANs — 172.16.0.0/16

Table 2: VLAN subnets, SVI addresses, and HSRP virtual gateways

ID	Name	Subnet	/pfx	MLS-1 SVI	MLS-2 SVI	HSRP VIP
10	SALES	172.16.10.0	/25	172.16.10.1	172.16.10.2	172.16.10.3
20	HR	172.16.20.0	/25	172.16.20.1	172.16.20.2	172.16.20.3
30	FINANCE	172.16.30.0	/25	172.16.30.1	172.16.30.2	172.16.30.3
40	ADMINS	172.16.40.0	/25	172.16.40.1	172.16.40.2	172.16.40.3
50	ICT	172.16.50.0	/25	172.16.50.1	172.16.50.2	172.16.50.3
60	SERVERS	172.16.60.0	/28	172.16.60.1	172.16.60.2	172.16.60.3
100	MGMT	172.16.99.0	/26	172.16.99.1	172.16.99.2	172.16.99.3

Subnet mask reference: /25 → 255.255.255.128 (126 usable hosts) /28 → 255.255.255.240 (14 usable) /26 → 255.255.255.192 (62 usable)

3.2 Core Point-to-Point Links — 10.0.0.0/24

Table 3: Routed /30 links between multilayer switches and edge routers

Link	Subnet	MLS port / IP	Router port / IP	Mask
MLS-1 ↔ R-1	10.0.0.0/30	Gi1/0/10 — 10.0.0.1	G0/2 — 10.0.0.2	255.255.255.252
MLS-1 ↔ R-2	10.0.0.8/30	Gi1/0/11 — 10.0.0.9	G0/0 — 10.0.0.10	255.255.255.252
MLS-2 ↔ R-1	10.0.0.4/30	Gi1/0/10 — 10.0.0.5	G0/0 — 10.0.0.6	255.255.255.252
MLS-2 ↔ R-2	10.0.0.12/30	Gi1/0/11 — 10.0.0.13	G0/2 — 10.0.0.14	255.255.255.252

3.3 ISP-Facing Addressing — RFC 5737 Documentation Ranges

Table 4: ISP simulation addressing

Link	Subnet	ISP port / IP	Connected device / IP	Mask
ISP-R ↔ R-1	198.51.100.0/30	G0/0 — 198.51.100.1	R-1 G0/1 — 198.51.100.2	255.255.255.252
ISP-R ↔ R-2	198.51.100.4/30	G0/1 — 198.51.100.5	R-2 G0/1 — 198.51.100.6	255.255.255.252
ISP public LAN	203.0.113.0/24	G0/2 — 203.0.113.1	HTTP/HTTPS Server — 203.0.113.10	255.255.255.0

4 Technologies Implemented

4.1 VLAN Segmentation and VTP

Seven VLANs are configured across the entire switching infrastructure, one per department plus a dedicated management VLAN (100). VLAN propagation is managed via VTP: both multilayer switches act as VTP servers, and all six access switches are configured as VTP clients within the same VTP domain. All inter-switch links use 802.1Q trunking with an explicit allowed VLAN list (10, 20, 30, 40, 50, 60, 100), preventing unintended VLAN leakage. Isolating each department into its own VLAN limits broadcast domain size and prevents lateral traffic between departments at Layer 2, improving both performance and security posture.

4.2 Inter-VLAN Routing via Multilayer Switching

IP routing is enabled on both MLS switches (`ip routing`). Each VLAN is associated with a Switch Virtual Interface (SVI) configured with a dedicated IP address, allowing Layer 3 forwarding between departmental subnets directly at the core layer without relying on

an external router. Using SVIs on the multilayer switches rather than a dedicated router-on-a-stick configuration eliminates an external device as a single point of failure and keeps inter-VLAN traffic within the campus switching fabric.

4.3 EtherChannel — LACP

A 3-link Port-Channel (Po10) is established between MLS-1 and MLS-2 using LACP in active mode. Physical interfaces Gi1/0/7, Gi1/0/8, and Gi1/0/9 are bundled on both switches. The resulting Port-Channel is configured as an 802.1Q trunk carrying all VLANs and is marked as a point-to-point link for Spanning Tree purposes. Three physical links were bundled rather than one to provide both increased throughput and link-level redundancy — a single cable failure does not degrade the inter-switch connection.

4.4 Spanning Tree — Rapid PVST+

Rapid PVST+ is deployed on all switches with PortFast enabled globally for access-facing ports to minimize convergence time for end-host links. Root bridge election is controlled per VLAN through explicit priority configuration:

- **MLS-1** is root bridge for VLANs 10–60 (priority 4096), anchoring all user and server traffic to the primary core switch during normal operation.
- **MLS-2** is root bridge for VLAN 100 (priority 4096), separating the management plane from user traffic.

Splitting root bridge responsibility between MLS-1 and MLS-2 distributes the Layer 2 forwarding load and ensures that a failure of either switch does not simultaneously disrupt both user traffic and management access.

4.5 HSRP — First Hop Redundancy

HSRP is configured on all SVIs. Each VLAN is assigned a virtual IP address that serves as the default gateway for all hosts in that VLAN. MLS-1 is the active gateway for VLANs 10–60 with priority 110 and preemption enabled. MLS-2 remains standby for those VLANs and is active for VLAN 100. The HSRP virtual IP is the only address configured as a default gateway on end hosts, which means a failover between MLS-1 and MLS-2 is completely transparent to users — no host reconfiguration is required.

4.6 DHCP Relay

A Packet Tracer DHCP server resides in VLAN 60 at address 172.16.60.4. Since DHCP broadcast requests cannot cross VLAN boundaries natively, the command `ip helper-address 172.16.60.4` is applied to the SVI of each user VLAN on both MLS switches, forwarding DHCP requests as directed unicast to the centralized server. Centralizing DHCP in VLAN 60 rather than running separate servers per VLAN simplifies address management and ensures consistent lease policies across all departments.

4.7 Static Routing and Edge Connectivity

Static routing is used throughout for deterministic behavior and straightforward troubleshooting. Each MLS switch has a primary default route toward one edge router and a floating backup default route (administrative distance 5) toward the other. Each edge router contains static routes toward 172.16.0.0/16 through both MLS switches. The ISP router holds static routes back to the campus through both R-1 and R-2. Static routing was chosen over a dynamic protocol because the topology is small and stable — the overhead and complexity of OSPF are not justified for two edge routers serving a single campus, and floating backup

routes provide adequate failover.

5 Port Mapping

Table 5: MLS-1 port connections

MLS-1 Port	Device	Remote Port	VLAN	Description
Gi1/0/1	SW7	Fa0/2	60	Trunk toward Servers switch (Floor 3)
Gi1/0/2	SW6	Fa0/2	50	Trunk toward ICT switch (Floor 3)
Gi1/0/3	SW5	Fa0/2	40	Trunk toward Admins switch (Floor 2)
Gi1/0/4	SW4	Fa0/2	30	Trunk toward Finance switch (Floor 2)
Gi1/0/5	SW3	Fa0/2	20	Trunk toward HR switch (Floor 1)
Gi1/0/6	SW2	Fa0/2	10	Trunk toward Sales switch (Floor 1)
Gi1/0/7–9	MLS-2	Gi1/0/7–9	All	EtherChannel Po10 (LACP)
Gi1/0/10	R-1	G0/2	—	Routed link — 10.0.0.0/30
Gi1/0/11	R-2	G0/0	—	Routed link — 10.0.0.8/30

Table 6: MLS-2 port connections

MLS-2 Port	Device	Remote Port	VLAN	Description
Gi1/0/1	SW2	Gi0/1	10	Trunk toward Sales switch (Floor 1)
Gi1/0/2	SW3	Gi0/1	20	Trunk toward HR switch (Floor 1)
Gi1/0/3	SW4	Gi0/1	30	Trunk toward Finance switch (Floor 2)
Gi1/0/4	SW5	Gi0/1	40	Trunk toward Admins switch (Floor 2)
Gi1/0/5	SW6	Gi0/1	50	Trunk toward ICT switch (Floor 3)
Gi1/0/6	SW7	Gi0/1	60	Trunk toward Servers switch (Floor 3)
Gi1/0/7–9	MLS-1	Gi1/0/7–9	All	EtherChannel Po10 (LACP)
Gi1/0/10	R-1	G0/0	—	Routed link — 10.0.0.4/30
Gi1/0/11	R-2	G0/2	—	Routed link — 10.0.0.12/30

6 Configuration Verification

The following outputs were captured from the configured devices and confirm that all implemented technologies are fully operational.

6.1 HSRP Status

MLS-1 is confirmed as the active HSRP gateway for VLANs 10–60 with priority 110 and preemption enabled. MLS-2 is active for VLAN 100 and standby for all others, consistent with the design intent.

```
MLS-1# show standby brief
Interface Grp Pri P State Active Standby Virtual IP
Vl10 10 110 P Active local 172.16.10.2 172.16.10.3
Vl20 20 110 P Active local 172.16.20.2 172.16.20.3
Vl30 30 110 P Active local 172.16.30.2 172.16.30.3
Vl40 40 110 P Active local 172.16.40.2 172.16.40.3
Vl50 50 110 P Active local 172.16.50.2 172.16.50.3
Vl60 60 110 P Active local 172.16.60.2 172.16.60.3
Vl100 100 100 Standby 172.16.99.2 local 172.16.99.3
```

```
MLS-2# show standby brief
Interface Grp Pri P State Active Standby Virtual IP
Vl10 10 100 P Standby 172.16.10.1 local 172.16.10.3
Vl20 20 100 P Standby 172.16.20.1 local 172.16.20.3
Vl30 30 100 P Standby 172.16.30.1 local 172.16.30.3
Vl40 40 100 P Standby 172.16.40.1 local 172.16.40.3
Vl50 50 100 P Standby 172.16.50.1 local 172.16.50.3
Vl60 60 100 P Standby 172.16.60.1 local 172.16.60.3
Vl100 100 100 Active local 172.16.99.1 172.16.99.3
```

6.2 EtherChannel Verification

Port-channel10 is active (SU flag) using LACP, with all three physical member links in a bundled state (P flag). No degraded or suspended members are present on either switch.

```
MLS-1# show etherchannel summary
Group Port-channel Protocol Ports
-----+-----+-----+-----
10 Po10(SU) LACP Gig1/0/7(P) Gig1/0/8(P) Gig1/0/9(P)
```

```
MLS-2# show etherchannel summary
Group Port-channel Protocol Ports
-----+-----+-----+-----
10 Po10(SU) LACP Gig1/0/7(P) Gig1/0/8(P) Gig1/0/9(P)
```

6.3 Interface Status — MLS-1

All seven SVIs are confirmed up/up with their correct IP addresses. The two routed uplinks (Gi1/0/10 toward R-1 and Gi1/0/11 toward R-2) are operational. VLAN 1 is administratively shut down as expected since it is not used.

```

MLS-1# show ip interface brief
Interface                IP-Address    OK? Method Status  Protocol
Port-channel10           unassigned    YES unset  up      up
GigabitEthernet1/0/1     unassigned    YES unset  up      up
GigabitEthernet1/0/2     unassigned    YES unset  up      up
GigabitEthernet1/0/3     unassigned    YES unset  up      up
GigabitEthernet1/0/4     unassigned    YES unset  up      up
GigabitEthernet1/0/5     unassigned    YES unset  up      up
GigabitEthernet1/0/6     unassigned    YES unset  up      up
GigabitEthernet1/0/7     unassigned    YES unset  up      up
GigabitEthernet1/0/8     unassigned    YES unset  up      up
GigabitEthernet1/0/9     unassigned    YES unset  up      up
GigabitEthernet1/0/10    10.0.0.1      YES NVRAM  up      up
GigabitEthernet1/0/11    10.0.0.9      YES NVRAM  up      up
Vlan1                    unassigned    YES NVRAM  admin down down
Vlan10                   172.16.10.1   YES NVRAM  up      up
Vlan20                   172.16.20.1   YES NVRAM  up      up
Vlan30                   172.16.30.1   YES NVRAM  up      up
Vlan40                   172.16.40.1   YES NVRAM  up      up
Vlan50                   172.16.50.1   YES NVRAM  up      up
Vlan60                   172.16.60.1   YES NVRAM  up      up
Vlan100                  172.16.99.1   YES NVRAM  up      up

```

6.4 VLAN Database

The following output confirms all seven VLANs are present and active on both multilayer switches with consistent naming, confirming successful VTP propagation across the domain.

```

MLS-1# show vlan brief
VLAN  Name      Status
----  -
1      default  active
10     SALES    active
20     hr       active
30     finance  active
40     admin    active
50     ict      active
60     servers  active
100    MNGMNT   active

```

```

MLS-2# show vlan brief
VLAN  Name      Status
----  -
1      default  active
10     SALES    active
20     hr       active
30     finance  active
40     admin    active
50     ict      active
60     servers  active
100    MNGMNT   active

```

6.5 Spanning Tree Verification

MLS-1 is confirmed as root bridge for VLAN 10 (priority 4106 = 4096 + sys-id-ext 10). All access-facing ports and Po10 are in the Designated Forwarding state.

```
MLS-1# show spanning-tree vlan 10
  Root ID    Priority    4106
             Address     0004.9AA1.8E41
             This bridge is the root

  Bridge ID  Priority    4106 (priority 4096 sys-id-ext 10)

Interface    Role Sts  Cost    Prio.Nbr  Type
-----
Po10         Desg FWD  3       128.29   P2p
Gi1/0/1      Desg FWD  19      128.1    P2p
Gi1/0/2      Desg FWD  19      128.2    P2p
Gi1/0/3      Desg FWD  19      128.3    P2p
Gi1/0/4      Desg FWD  19      128.4    P2p
Gi1/0/5      Desg FWD  19      128.5    P2p
Gi1/0/6      Desg FWD  19      128.6    P2p
```

6.6 Trunk Verification

All six access-facing ports and Port-channel10 on MLS-1 are confirmed as 802.1Q trunks, with VLANs 10, 20, 30, 40, 50, 60, and 100 active and forwarding.

```
MLS-1# show interfaces trunk
Port      Mode  Encapsulation  Status  Native vlan
Po10      on    802.1q         trunking  1
Gi1/0/1   on    802.1q         trunking  1
Gi1/0/2   on    802.1q         trunking  1
Gi1/0/3   on    802.1q         trunking  1
Gi1/0/4   on    802.1q         trunking  1
Gi1/0/5   on    802.1q         trunking  1
Gi1/0/6   on    802.1q         trunking  1

Port      Vlans in spanning tree forwarding state and not pruned
Po10      10,20,30,40,50,60,100
Gi1/0/1   10,20,30,40,50,60,100
```

6.7 Routing Verification — MLS-1

All seven VLAN subnets appear as directly connected. The primary default route is via R-1 (10.0.0.2). The floating backup default route via R-2 activates automatically if the primary path becomes unreachable.

```
MLS-1# show ip route
Gateway of last resort is 10.0.0.2 to network 0.0.0.0

C    10.0.0.0/30   directly connected, GigabitEthernet1/0/10
S    10.0.0.4/30   [1/0] via 10.0.0.2
C    10.0.0.8/30   directly connected, GigabitEthernet1/0/11
S    10.0.0.12/30  [1/0] via 10.0.0.10
C    172.16.10.0/25 directly connected, Vlan10
```

```
C 172.16.20.0/25 directly connected, Vlan20
C 172.16.30.0/25 directly connected, Vlan30
C 172.16.40.0/25 directly connected, Vlan40
C 172.16.50.0/25 directly connected, Vlan50
C 172.16.60.0/28 directly connected, Vlan60
C 172.16.99.0/26 directly connected, Vlan100
S* 0.0.0.0/0 [1/0] via 10.0.0.2
```

6.8 Routing Verification — R-1

R-1 shows correct static routes toward the campus range via both MLS switches and a default route toward the ISP, confirming bidirectional routing between the campus and the simulated public segment.

```
R-1# show ip route
C 10.0.0.0/30 directly connected, GigabitEthernet0/2
C 10.0.0.4/30 directly connected, GigabitEthernet0/0
S 10.0.0.8/30 [1/0] via 10.0.0.1
S 10.0.0.12/30 [1/0] via 10.0.0.5
S 172.16.0.0/16 [1/0] via 10.0.0.5
[1/0] via 10.0.0.1
C 198.51.100.0/30 directly connected, GigabitEthernet0/1
S* 0.0.0.0/0 [1/0] via 198.51.100.1
```

6.9 DHCP End-to-End Verification

The following output was captured from PC1, a host in VLAN 10 (Sales). It confirms that the full DHCP relay chain is operational.

```
PC1> ipconfig

FastEthernet0 Connection:
    IPv4 Address . . . : 172.16.10.5
    Subnet Mask . . . : 255.255.255.128
    Default Gateway . . : 172.16.10.3
```

What this output confirms:

- The Packet Tracer DHCP server in VLAN 60 is reachable from VLAN 10 hosts via the `ip helper-address` relay configured on the MLS SVIs — DHCP requests successfully crossed the VLAN boundary.
- The assigned address (172.16.10.5) falls within the correct VLAN 10 subnet (172.16.10.0/25, usable range .1–.126).
- The default gateway distributed to the host is 172.16.10.3, which is the HSRP virtual IP — not a physical SVI address. This means gateway failover is fully transparent to end hosts: if MLS-1 fails, MLS-2 takes over the virtual IP with no reconfiguration required on PC1.

7 Known Limitations

- **Simulator constraints:** the project runs entirely within Cisco Packet Tracer. Protocol behaviors, timing, and feature support are subject to simulator limitations and may not fully reflect real hardware behavior.
- **Static routing only:** no dynamic routing protocol is implemented. OSPF or EIGRP were intentionally excluded because the topology is a single-campus, single-company design where static routing with floating backup routes provides adequate and controllable redundancy. A multi-site or multi-company scenario would justify dynamic routing.
- **No firewall policies:** pfSense and FortiGate appliances are referenced in the design but are not configured in this phase. Inter-VLAN filtering, perimeter rules, and zone-based policies are planned as a dedicated security project.
- **No ACL on management VLAN:** VLAN 100 is present and operational, but no access control list restricts which hosts can reach management interfaces. This is a known security gap acknowledged for the next phase.
- **ISP simulation only:** the external connectivity segment uses RFC 5737 documentation addresses and a Packet Tracer server. This does not represent real Internet connectivity.